

Track A item 1: does the $q = i$ reflected-contour
lemma extend to $N = 6$?

STATUS: OPEN, with a sharper diagnosis than “ $T < 0$ ”

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Abstract

At $N = 6$ ($a/N = 1/6, 5/6$) the P1f/P1i term-by-term method for the poles of U, V (accumulation at $\pm\sqrt{e^{2\pi i(\pm 1/6)}}$, standing on `thm:model/thm:RJ` as always in this programme) is documented (`P1i_lemma.tex`, § “ $N = 6$: exactly why it is not closed”) as structurally unclosable by the crude landscape method: the tie height is $T = -0.10878 < 0$ while the head term $b_0 = 1$ sits at height 0 (indeed $H_6 - T \geq 0.1548$), so no term-by-term cutoff bound of the P1f/P1h shape can ever work at $N = 6$, regardless of constants. `P1i_lemma.tex` and `paper2a.tex` (Remark `rem:arcobstruction`) both flag that the only known route around an analogous obstruction — met and resolved at $q = i$ ($N = 4$, $T = -0.0707$) by `prop:qi` — would need `lem:qi-bounds(ii)` “generalised from the two residue classes used at $q = i$ to six,” and call this open without saying why it is not simply a relabelling exercise.

This note reports an attempt to carry out that generalisation. **Result: it does not go through as stated, and the reason is not a restatement of $T < 0$.** The tie at $N = 6$ is a clean two-saddle (“pairwise”) tie exactly as at $N = 4$ — so the fear that $6 = 2 \cdot 3$ ’s extra divisor forces a three-way tie, breaking the two-term Rouché argument, is checked and is *false*. The phase function $\Phi(x) = xL - x^2 - \text{Li}_2(e^{-2Nx})/N^2 + \pi^2/(6N^2)$ and the Jacobi-triple-product/Poisson-summation core of `lem:qi-bounds(ii)` are themselves fully N -generic and carry over to $N = 6$ with only bookkeeping changes (mod-6 residues in place of mod-4, and the pre-existing mod-2 even/odd branch split nests inside mod-6 exactly as it nests inside mod-4, since $2 \mid 6$ just as $2 \mid 4$). **The actual obstruction is that the *ray lemma* underlying every part of `lem:qi-bounds` — the tool that produces the separation constants δ , $d(x)$, $d'(x)$ that make the whole $\text{Re } x \leq 0$ continuation rigorous — is proved only under $\sin \theta \neq 0$ (the tie-ray angle), via the explicit statement**

“the argument of w turns at rate $\sin \theta$.” At $q = i$ this holds: $\theta_i = \arctan(\pi/(6 \log 2)) \neq 0$. At $N = 6$ the certified tie-ray angle is exactly $\theta^* = 0$ (P1i_lemma.tex, §??-data), i.e. the ray is real and $w = ce^{-v}$ does not rotate away from resonance as $\rho \rightarrow \infty$ along it. As currently proved, the ray lemma’s proof gives no guaranteed lower bound at $\theta = 0$ without a modulus-only rescue, which we did not check (Reviewer AK, post-review correction: this is weaker than saying the lemma cannot hold at $N = 6$ — see §5). This is a genuinely different, and sharper, statement than “ $T < 0$ ”: it says the one analytic tool capable of getting around $T < 0$ is itself degenerate at exactly the angle where $N = 6$ needs it.

1 What was read

paper2a.tex (prop:qi, lem:qi-bounds, and § “the middle arc”/Remarks rem:arcobstruction, rem:polesdensity); general/P1i/P1i_lemma.tex (§ “ $N = 6$: exactly why it is not closed,” and the small- N table in the proof of the landscape certificate); general/P1f/P1f_lemma.tex (the N -generic phase function Φ and § “head”); qi_bounds.tex, certify_landscape.i.py, certify_landscape.S.i.py, factors.i.py (the $q = i$ certificates, read for their role, not re-run — see §5). No paper .tex file was edited.

2 The $N = 4$ mechanism, precisely

2.1 Why $T < 0$ blocks the crude method

The P1f/P1h machinery decomposes B (equivalently $1 - \Sigma_1$) as a head H (finitely many low coefficients b_k , dominated by $b_0 = 1$) plus a sum of saddle-mode integrals plus kernel remainders. The non-head pieces decay like $e^{(T-\eta)/r}$ for the certified tie height T ; the head is controlled separately by lem:head, which needs $N \geq 16$ and gives $H_{\text{head}} \leq 0.0167 + r$. When $T < 0$ and the head sits at height 0 (or, at $N = 6$, strictly above T by ≥ 0.1548), *no* choice of cutoff or constants can make a term-by-term bound $|H| \leq e^{(T-\eta)/r}$ hold, because H genuinely does not decay — $b_0 = 1$ is a constant. This is the mechanism common to $q = i$ and $N = 6$, and it is a hard wall for *this method*, not a numerical accident.

2.2 The resolution at $q = i$: eliminate the head, don't bound it

`prop:qi`'s proof does not find a sharper bound on H . It changes the decomposition so that no head term appears at all: Step 1 starts the contour split at a *negative* half-integer $s_c < 0$ (with x_c near $-\frac{\pi}{8}(\cot \theta_i + i)$), so that $\sum_{k \geq 0} F(k)$ is written via $K_- = (1 - e^{-2\pi i s})^{-1}$, $K_+ = e^{2\pi i s}(1 - e^{2\pi i s})^{-1}$ over paths $\Gamma_{\mp}$ leaving s_c , and the entire term is moved to a path Π_0 through the *saddle* x_0 rather than through the would-be head index $s = 0$. Concretely, b_0 is folded into the analytic integral $J_0 = \int_{\Pi_0} F ds$, which the Laplace-method Step 3 controls via the saddle height V_0 , not via a term-by-term coefficient bound. Making this legitimate requires that F (the interpolant, built from f_e, f_o) be controlled uniformly on paths that cross into $\operatorname{Re} x \leq 0$ (the paths $\Gamma_{\mp}$ and Π_0 pass through $\operatorname{Im} x = \pm\pi/8$ and effectively sample F on both sides of $\operatorname{Re} x = 0$). That control is exactly `lem:qi-bounds(ii)`: a bound on $\log |f(x)|$ for $\operatorname{Re} x \leq 0$, obtained by writing $(z; P)_{\infty} = \vartheta(z)/(P/z; P)_{\infty}$, expanding ϑ via the Jacobi triple product, Poisson-summing over $k \in \mathbb{Z}$ (the resulting Gaussian sum in k is a concave quadratic, so it is dominated, up to the factor $3 + 3e^{-\pi^2 \cos \theta/r}$, by its two largest terms near m^*), and bounding the remaining $(P/z_j; P)_{\infty}$ factor by the *ray lemma* with separation constant $\delta \geq d'(x)$.

2.3 Is the two-tied-saddle structure special to $N = 4$?

No: the tie itself (two saddle modes x_0, x_- of equal height along $\theta = \theta_i$, coming from the quasi-period shift $\Phi_i(x - i\pi/2) = \Phi_i(x) - \frac{i\pi L_w}{2} + i\pi x + \frac{\pi^2}{4}$, itself a special case at $N = 4$ of the general fact that $\operatorname{Li}_2(e^{-2Nx})$ has exact period $i\pi/N$ in x , so shifting by a half-period picks up a computable affine correction) is the same kind of object as the general P1f/P1h “two consecutive admissible modes $n, n+s$ tie along an explicit tie ray” statement used for all N . `P1i_lemma.tex` confirms this directly for $N = 6$: the certified tie pair is $(-3, -1)$, i.e. a clean *pairwise* tie, not a three-way or six-way tie. **So the hypothesis floated in the task description — that $6 = 2 \cdot 3$'s extra proper divisor 3 might force a multi-way (not pairwise) tie that breaks the two-term Rouché argument $g_0(u) = 1 - e^{\Delta V(u-u_j)}$ — is checked against the existing certificate and is false.** The Rouché step itself would carry over unchanged in shape.

Likewise the phase function $\Phi(x) = xL - x^2 - \operatorname{Li}_2(e^{-2Nx})/N^2 + \pi^2/(6N^2)$ (`P1f_lemma.tex`) and the Pochhammer factorisation $(q; q)_{\infty} = \prod_{i=1}^N (\zeta^i p^i; p^N)_{\infty}$ used throughout the programme are already fully N -generic; nothing in

them privileges $N = 4$. The pre-existing mod-2 split into even/odd branches f_e, f_o (used for *every* N in this programme, to handle b_{2s} vs. b_{2s+1}) nests inside a further mod-6 residue decomposition of the Pochhammer product exactly as it nests inside mod-4, because $2 \mid 6$ just as $2 \mid 4$: there is no clash between the mod-2 branch structure and a mod-6 theta decomposition analogous to the clash one might fear from 6 having two distinct prime factors. **So the “ $4 = 2^2$ vs. $6 = 2 \cdot 3$ ” framing in the task description, taken as “does 6 having a genuinely different residue structure break the nesting,” is also not the true obstruction:** the nesting survives.

3 The actual obstruction found

The one place where $N = 4$ is used in `lem:qi-bounds`’s *proof*, not merely in its statement, is the *ray lemma* that opens the proof of the lemma. Quoting its derivation: for $G(v) = \log(1 - ce^{-v})$ along a ray $v_0 + e^{i\theta}\mathbb{R}_{\geq 0}$, the separation constant δ used to bound $|G'| \leq 1/\delta$ and $\int |G''| \leq 1/(\delta^2 \cos \theta)$ is obtained because “ $w = ce^{-v}$ has $|w| \leq e^{-a' - \rho \cos \theta}$ and $|1 - w| \geq \max(1 - |w|, \sigma(\arg w))$, and *the argument of w turns at rate $\sin \theta$* ; this gives $\delta \geq \mathcal{D}(a', \varphi)$.” In words: as ρ increases along the ray, w both shrinks in modulus (rate $\cos \theta$) and rotates in argument (rate $\sin \theta$); the rotation is what guarantees that w does not sit exactly on top of 1 (or any other resonance point) for an extended stretch of the ray, which is what the quantitative lower bound $\mathcal{D}(a', \varphi)$ on $|1 - w|$ needs. Every part of `lem:qi-bounds` — (i)’s E -bound, (ii)’s $M'(x)$, $K'(x)$, and even (iv)’s kernel bound — is built by invoking this ray lemma (or its δ', D_n variants) along rays with $\theta = \theta_i \neq 0$.

At $N = 6$ the certified tie-ray angle is $\theta^* = 0$ exactly (`P1i_lemma.tex`, § “ $N = 6$: exactly why it is not closed”: “ $c = 2e^{2\pi i/3}$, $\ell = \log 2$, $\theta^* = 0$, tie pair $(-3, -1)$ ”). At $\theta = 0$, $\sin \theta = 0$: the ray is exactly the positive real axis in the relevant coordinate, $w = ce^{-v}$ does not rotate at all as $\rho \rightarrow \infty$, and the argument-turning mechanism that produces $\mathcal{D}(a', \varphi)$ has no content. If $\arg w_0$ happens to sit near a resonance ($\arg w_0 \approx 0 \pmod{2\pi}$), i.e. c itself near a positive real axis crossing of the relevant singular set), the ray lemma’s proof, as written, gives no lower bound on $|1 - w|$ along the whole ray, and hence no bound on $\log |f(x)|$ of the shape `lem:qi-bounds`(i)/(ii) at that θ . This would need to be checked case-by-case for the actual c ’s occurring in the $N = 6$ decomposition (there may be enough residual modulus decay, $\cos \theta = 1$ being the *best* possible rate for the modulus half of the estimate, to compensate for the loss of the rotation half in some or all of the needed windows) — but that is exactly the missing derivation, not a bookkeeping

relabelling from 4 to 6.

Why this is a sharper diagnosis than “ $T < 0$ ”: $T < 0$ says the crude term-by-term method cannot possibly work at $N = 6$, which was already known (`P1i_lemma.tex`). It does *not* say why the one alternative method on record (the $q = i$ reflected-contour trick) fails to transfer. The finding here is that the alternative method’s core analytic tool, the ray lemma, was proved under a nondegeneracy hypothesis ($\sin \theta \neq 0$) that happens to hold at $q = i$ ’s own tie angle but fails at $N = 6$ ’s own tie angle — a fact about the *geometry of $N = 6$ ’s specific tie ray*, not about the sign of T , and not about 6’s divisor structure either (the two hypotheses suggested in the task prompt were checked and ruled out in §3 above).

Remark 1 ($\theta^* = 0$ is not universally fatal). The certificate table in `P1i_lemma.tex` also lists $\theta^* = 0$ at $a/N = 3/10$, one of the 52 members where P1f’s *general* (non- $q-i$ -style) method *does* close successfully. This does not contradict the diagnosis above: $N = 10 \geq 16$ fails the P1f hypothesis too in principle, but that particular member is handled by `P1i_lemma.tex`’s per-case Arb landscape certificate (`P1i_landscape_cert.py`), which does not go through `lem:qi-bounds` or its ray lemma at all — it replaces the analytic Lemmas LS/LSp/head of P1f with a direct numerical certificate of the landscape inequalities, and never needs a head-elimination trick because at $3/10$ the head does not exceed the tie height. So $\theta^* = 0$ being survivable for the general P1f certificate machinery does not show it is survivable for the *ray lemma* specifically, which is the one piece $N = 6$ actually needs (since $N = 6$ ’s head genuinely beats its tie height, unlike $3/10$ ’s).

4 No certificate written

No Arb/python-flint certificate was written for $N = 6$ beyond confirming (by reading, not by a fresh computation) the tie-pair and $\theta^* = 0$ data already printed in `P1i_lemma.tex`. Writing a certificate would require first resolving §3: either (a) proving a $\theta = 0$ -degenerate replacement for the ray lemma (plausibly using pure modulus decay $|w| \leq e^{-a'-\rho}$ alone, without the rotation term, if the relevant c ’s at $N = 6$ stay far enough from $\arg = 0$ resonance in modulus alone — untested here), or (b) finding a different decomposition of B at $N = 6$ whose tie ray is not exactly $\theta = 0$ (not attempted here; it is not clear the tie ray is a free choice, since it is pinned by $\ell = \log 2$ and the specific $N = 6$, $a = 1$ data, the same way θ_i is pinned at $N = 4$). Neither was carried out. This note stops at the diagnosis in §3 and reports it as OPEN.

5 Weakest points, stated honestly

1. **The ray-lemma obstruction is a diagnosis, not a proof of impossibility.** I have not shown that no $\theta = 0$ version of the ray lemma can be proved; I have shown that the *existing* proof uses $\sin \theta \neq 0$ essentially, and that $N = 6$'s certified tie ray sits exactly at $\theta = 0$. It is plausible a modulus-only argument rescues the needed bound at $N = 6$'s specific $c = 2e^{2\pi i/3}$ -derived resonance points; I did not attempt to check this case-by-case (it would require identifying every c_j , $j = 1, \dots, 6$, occurring in the $N = 6$ Pochhammer/theta decomposition, and checking whether any lies close enough to a positive-real resonance to make the missing rotation term actually matter). This is squarely the open mathematical content flagged by `P1i_lemma.tex`, now with a specific, checkable target (the six c_j 's and their distance from $\arg = 0$) rather than a vague “generalise the lemma.”
2. **I did not re-derive the general mod- N Poisson/theta bookkeeping in full**, i.e. I did not write out the $N = 6$ analogues of $\Upsilon(x)$, $M'(x)$, $K'(x)$ term by term. I argued from the structure of the derivation (which is visibly N -generic apart from the ray lemma) that this bookkeeping step is mechanical, but “mechanical” claims in an eight-line proof with several explicit numeric constants (e.g. $\frac{73}{12}$, $\frac{4}{3}$) deserve suspicion until actually carried out; I have not carried it out.
3. **The claim that the pairwise-tie and mod-2/mod-6 nesting concerns are non-issues rests on reading `P1i_lemma.tex`'s printed tie-pair $(-3, -1)$ and on the general divisibility argument in §3, not on an independent re-derivation of the $N = 6$ saddle set from scratch.** If `P1i_lemma.tex` has an error in its tie-pair computation (I have no reason to think so, but did not re-derive it), this part of the analysis would need revisiting.
4. **No numerics were run.** Per the task's own framing this is by design (“this requires genuinely new mathematical work, not more numerics”), but it means the one thing that could most cheaply sharpen §3 into either a fix or a firm no-go — computing the six c_j 's for $N = 6$ and their distances from $\arg = 0$ resonance, which is a five-minute Arb computation, not a search — was deliberately left undone here and is the natural next step for anyone picking this up.

6 Conclusion

OPEN. The reflected-contour ($q = i$ -style) route does not obviously extend to $N = 6$, and the reason is not a restatement of $T < 0$, nor the multi-way-tie fear, nor a divisor-structure clash in the mod- N residue bookkeeping (both checked and ruled out). The reason found is that the ray lemma underlying **lem:qi-bounds** is proved under $\sin(\text{tie angle}) \neq 0$, and $N = 6$'s own certified tie angle is exactly 0. Whether a degenerate-angle version of the ray lemma can be proved, or whether it fails outright for the specific resonance data at $N = 6$, is not resolved here.